

The Trinity Study and Portfolio Success Rates

Following Bengen, another of the classic studies in the field of financial and retirement planning is the “Trinity study,” a nickname for the article “Retirement Spending: Choosing a Sustainable Withdrawal Rate,” by Philip L. Cooley, Carl M. Hubbard, and Daniel T. Walz (all professors at Trinity University in Texas). It appeared in the February 1998 issue of the *Journal of the American Association of Individual Investors*.

This research followed the same methodology used by William Bengen in his 1994 article. What was different in the Trinity study was the shift in emphasis away from William Bengen’s SAFEMAX, or the highest withdrawal rate possible in the worst-case scenario from history, toward the idea of “portfolio success rates.” An unfortunate side effect of this redirection is that the meaning of these portfolio success rates has been widely misinterpreted by the public.

There was one important difference in the assumptions between William Bengen’s work and the Trinity study, which I think furthered some of the misunderstandings about the latter study. This difference regards the choice of bond indexes. While Mr. Bengen’s original research combined the S&P 500 index with intermediate-term government bond returns, the Trinity study used long-term high-grade corporate bond returns instead. The different choice for bonds explains why the worst-case scenario for Mr. Bengen (his SAFEMAX) was a withdrawal rate of 4.15 percent, but the original Trinity study found that a 4 percent withdrawal rate only had a 95 percent success rate. With more volatile corporate bonds, the sustainable withdrawal rate dipped slightly below 4 percent in 1965 and 1966. This led people to hear that the 4 percent rule has a 95 percent chance for success, though that is true only in the historical data. It does not imply that today’s retirees will enjoy the same chance for success. Nonetheless, a 95 percent chance for success has a special meaning in statistics about providing a good level of confidence, so it easy to misunderstand this point. The 4 percent rule became further entrenched.